Doubles Strategy: The ENTER Duplicate Key

Number Racks & Fact Strategies • Standard: CCSS.MATH.1.OA.C.6

BOOK STORY OPENER: Two Hands, One Workbench: Why 5+5 Bead Rows Mirror the Stack & Instant [ENTER] [+] Doubles

Designed in the Netherlands by math educator Adrian Treffers, the Rekenrek ('calculating rack') splits each row into 5 red and 5 white beads—mirroring the 5 fingers on your left and right hands. In Chapters 1 & 2 of From Blocks to the Stack, pressing [ENTER] clones Shelf X onto Shelf Y just like matching the bottom bead row to the top row for instant doubles!

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 1: Meet Your Math Workbench & Ch. 2: What Happens When You Press [ENTER]?

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Two-Row Number Rack (Rekenrek)

Push 6 beads on the top row (5 red, 1 white). Push 6 beads on the bottom row.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Observe the matching rows. $6 + 6 = (5 + 5) + (1 + 1) = 10 + 2 = 12$.

3. CALCULATOR ACTION (SC-6 KEYCAPS & STACK TRACE)

Keystroke Flow: [6] [ENTER] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

ENTER creates an instant identical copy of register X in register Y. This perfectly represents a physical double.

Core Investigation Problem:

Push 7 beads on each row of the Number Rack. Use [ENTER] [+] to double it on the calculator.

Near-Doubles: Doubles Plus One

Fact Fluency & Near Doubles • Standard: CCSS.MATH.2.OA.B.2

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Push 7 beads on top row (5 red, 2 white). Push 8 beads on bottom row (5 red, 3 white). Prediction: 7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[7] [ENTER] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[7]	7	Enter base number 7
[ENTER]	7	Duplicate 7
[+]	14	Double: 7 + 7 = 14
[1]	1	Near-double difference (1)
[+]	15	14 + 1 = 15!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Retyping the number to double it (e.g., [7] [ENTER] [7] [+]) instead of recognizing that [ENTER] duplicates Register X into Register Y.

Hands-On Block Remediation: Slide 7 beads on both rows of the Number Rack simultaneously, then press [7] [ENTER] and point to the screen: both Y and X already hold 7, so pressing [+] doubles it immediately.

Misconception 2: Confusing '3 groups of 4' with '3 + 4' when transitioning from repeated addition to multiplication.

Hands-On Block Remediation: Build 3 rows of 4 beads on the rack; have the student key [4] [ENTER] [4] [+] [4] [+] = 12 first, then compare directly with [4] [ENTER] [3] [×] = 12.

4. EXTENSION MATHEMATICAL INSIGHT

Near-doubles build upon doubles fluency. The calculator executes the base double, then accumulates the offset.

Number Racks (Rekenrek): 5-Structure & Doubles

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: NUMBER RACK & ENTER DUPLICATE RULES

- Each row has 5 red beads and 5 white beads.
- Pushing beads to the left counts them. 5 red + 5 white = 10 per row.
- Doubles shortcut: Enter a number, press [ENTER] to duplicate, then press [+]!

Key Sequence: 6 ENTER +

Problem 1: Calculate double 8 using [ENTER] [+].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 2: Solve near-double 8 + 9: double 8 then add 1.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 3: Calculate double 9 using [ENTER] [+].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 4: Solve 5 + 6 using doubles plus one.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Unit 04: Number Racks (Rekenrek): 5-Structure & Doubles — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 1–2

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: Push 7 beads on each row of the Number Rack. Use [ENTER] [+] to double it on the calculator.

Core Solution: 14 beads. Keystrokes: [7] [ENTER] [+]

Core Check Keys: 7 ENTER +

Extension Challenge: Use the near-doubles strategy to solve 6 + 7 on the calculator.

Extension Solution: 13. Keystrokes: [6] [ENTER] [+] [1] [+]

Ext. Check Keys: 6 ENTER + 1 +

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Calculate double 8 using [ENTER] [+].

Expected Answer: 16

SC-6 Verification Keys: 8 ENTER +

Problem 2: Solve near-double 8 + 9: double 8 then add 1.

Expected Answer: 17

SC-6 Verification Keys: 8 ENTER + 1 +

Problem 3: Calculate double 9 using [ENTER] [+].

Expected Answer: 18

SC-6 Verification Keys: 9 ENTER +

Problem 4: Solve 5 + 6 using doubles plus one.

Expected Answer: 11

SC-6 Verification Keys: 5 ENTER + 1 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.